

**UNIVERSITY OF BRISTOL**

**Summer Exam Period 2023 Examination Period**

**FACULTY OF ENGINEERING**

**Second Year Examination for the Degree of  
Bachelor of Science and Master of Engineering**

**COMS20011  
Data-Driven Computer Science**

**TIME ALLOWED:  
2 Hours**

This paper contains *20* questions.

Each question has exactly *one* correct answer.

All answers will be used for assessment.

The maximum for this paper is *100 marks*.

You may use a calculator.

Calculators must be non-programmable.

The exam is closed-book (so no additional materials are allowed).

You may write workings out on the exam paper, and blank pages are provided at the end for this purpose. These workings out will not be collected or marked. You must enter your answers on the provided answer sheet only.

**Other Instructions:**

**TURN OVER ONLY WHEN TOLD TO START WRITING**

**Q1.** Consider the pixel values of a small image below:

|    |   |   |    |   |   |
|----|---|---|----|---|---|
| 10 | 2 | 2 | 2  | 2 | 2 |
| 10 | 2 | 1 | 14 | 2 | 2 |
| 10 | 2 | 2 | 13 | 0 | 2 |
| 10 | 2 | 1 | 15 | 2 | 2 |
| 10 | 2 | 2 | 15 | 1 | 2 |
| 10 | 2 | 2 | 2  | 2 | 2 |

Using the position of the pixel with value 13 as the centre pixel to be convolved, apply the following convolution filter once using 8-connectivity and once using 4-connectivity:

$$\begin{pmatrix} -1 & 1 & -1 \\ -1 & 3 & -1 \\ -1 & 1 & -1 \end{pmatrix}$$

Which of the options below is the correct answer for the new pixel value in each connectivity case?

- A. For 8-connectivity it is 76, and for 4-connectivity it is 70
- B. For 8-connectivity it is 60, and for 4-connectivity it is 66
- C. For 8-connectivity it is 60, and for 4-connectivity it is 76
- D. For 8-connectivity it is 66, and for 4-connectivity it is 60
- E. For 8-connectivity it is 70, and for 4-connectivity it is 76

[6 marks]

**Q2.** REDACTED

**Q3.** Naive Bayes Classifier - The table below shows the probability of certain words from amongst a large selection of spam and not spam emails received at a university. The occurrence of the words and their probabilities are independent of each other.

| Word    | $p(word spam)$ | $p(word ¬spam)$ |
|---------|----------------|-----------------|
| Ink     | 0.80           | 0.30            |
| Term    | 0.02           | 0.93            |
| Summer  | 0.40           | 0.65            |
| Printer | 0.18           | 0.75            |
| Bulk    | 0.70           | 0.10            |

Making a Naive Bayes assumption, compute the probability of sentence S1 below being spam and the probability of sentence S2 below not being spam:

(cont.)

*S1- Buy printer ink in bulk at prices seen last Summer.*

*S2- The Summer term notes are by the printer.*

Choose the correct option for  $P(S1|spam)$  and  $P(S2|not\ spam)$ :

- A.  $P(S1|spam) = 0.0403$  and  $P(S2|not\ spam) = 0.0146$
- B.  $P(S1|spam) = 0.0146$  and  $P(S2|not\ spam) = 0.4534$
- C.  $P(S1|spam) = 0.0403$  and  $P(S2|not\ spam) = 0.0014$
- D.  $P(S1|spam) = 0.0146$  and  $P(S2|not\ spam) = 0.4095$
- E.  $P(S1|spam) = 0.0403$  and  $P(S2|not\ spam) = 0.4534$

[6 marks]

**Q4.** For a digitised sample acquired using 4Hz sampling and 4 quantisation levels, the following file has been provided:

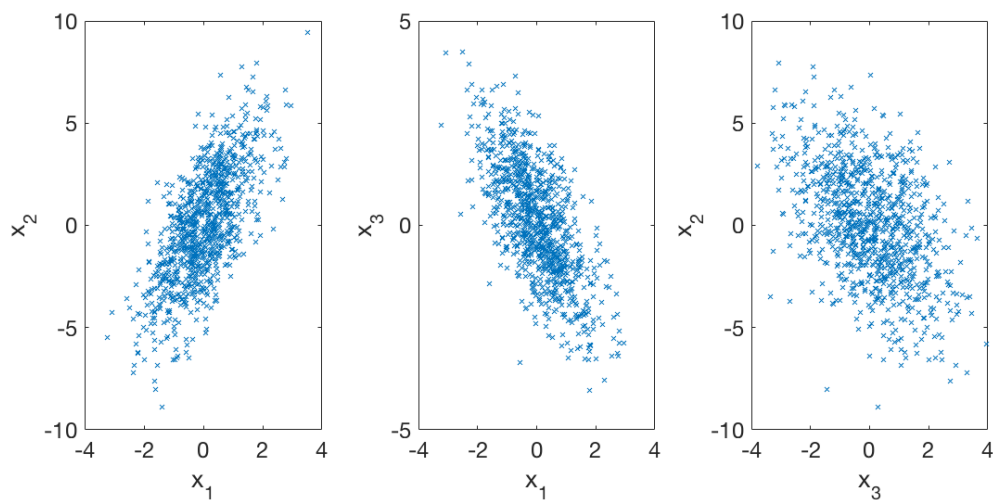
001010110101001001101011101010

The first sample collected after the first second of recording has passed is equal to:

- A. 0110
- B. 1011
- C. 01
- D. 10
- E. 0010

[3 marks]

**Q5.** For three-dimensional data  $X = (x_1, x_2, x_3)$ , we plot each variable against the other as shown below:



Given these plots, determine which of the following is a reasonable estimate of the covariance matrix  $\Sigma$  of dataset  $X$ ?

A.  $\Sigma = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 7 & -2 \\ -1 & -2 & 2 \end{bmatrix}$

B.  $\Sigma = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ -1 & -2 & 7 \end{bmatrix}$

C.  $\Sigma = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 1 & 2 \\ 1 & 2 & 7 \end{bmatrix}$

D.  $\Sigma = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 7 & -2 \\ -1 & -2 & 2 \end{bmatrix}$

E.  $\Sigma = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 7 & -2 \\ -1 & -2 & 0 \end{bmatrix}$

[5 marks]

**Q6.** Which of the following 2D matrices are NOT separable? Ignore normalisation factors which are not stated here.

$$M_1 = \begin{pmatrix} -1 & 3 & -1 \\ -3 & 9 & -3 \\ 1 & -3 & 1 \end{pmatrix}$$

$$M_2 = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ -1 & 0 & 1 \end{pmatrix}$$

$$M_3 = \begin{pmatrix} -1 & 4 & -1 \\ -1 & 8 & -1 \\ -1 & 4 & -1 \end{pmatrix}$$

$$M_4 = \begin{pmatrix} 1 & 2 & -1 & 2 & 4 \\ 2 & 4 & -2 & 4 & 8 \\ -1 & -2 & 1 & -2 & -4 \\ 2 & 4 & -2 & 4 & 8 \\ 4 & 8 & -4 & 8 & 16 \end{pmatrix}$$

$$M_5 = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 2 & 1 & 1 & -2 \\ 1 & 0 & 0 & -1 \\ 2 & 1 & 1 & -2 \end{pmatrix}$$

Choose the correct option:

- A.  $M_1$  and  $M_5$
- B.  $M_2$  and  $M_3$  and  $M_5$
- C.  $M_1$  and  $M_4$
- D.  $M_2$  and  $M_3$  and  $M_4$
- E.  $M_3$  and  $M_5$

[5 marks]

**Q7.** Two eigenvalues of the matrix below are 1 and 8:

$$\begin{bmatrix} 1 & -2 & 0 & 5 \\ 0 & 7 & 1 & 5 \\ 0 & 4 & 4 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

What are the other two eigenvalues?

- A. 3 and 4
- B. 4 and 2
- C. 2 and 3
- D. 1 and 3
- E. 0 and 4

[4 marks]

**Q8.** REDACTED

**Q9.** The eigenvalues of a dataset are: [26.0, 16.0, 13.0, 5.0, 4.0, 3.0, 1.95, 0.85, 0.60]. Approximately what variance in the dataset do the first 4 eigenvalues represent?

(cont.)

- A. 91.2%
- B. 88.6%
- C. 93.3%
- D. 77.3%
- E. 85.2%

[5 marks]

**Q10.** What is the minimum Edit Distance between the words "Sunday" and "Saturday"?

- A. 4
- B. 6
- C. 7
- D. 3
- E. 2

[3 marks]

**Q11.** Which of these is NOT a potential cause of overfitting?

- A. Choosing a function class that is too complex
- B. Choosing a function class that is too simple
- C. No regularisation.
- D. Too few datapoints.
- E. Datapoints only cover a small region in the input space.

[5 marks]

**Q12.** Which of these statements about cross-validation is FALSE:

- A. Cross-validation can be used to assess overfitting.
- B. Cross-validation reports performance on the training data used to fit the function.
- C. Cross-validation can be used to choose the function class.
- D. Cross-validation can be used to choose the amount of regularisation.
- E. Cross-validation can be computationally expensive if we have more than one or two hyperparameters.

[5 marks]

**Q13.** Which of these statements about the logarithm and its use in data-science is FALSE:

- A. The logarithm converts products into sums, i.e.  $\log ab = \log a + \log b$ .
- B. The logarithm converts powers into products, i.e.  $\log a^b = b \log a$ .

(cont.)

- C. The gradient of the logarithm is  $\frac{\partial \log p}{\partial p} = 2p^{-1}$ .
- D. Using log-probabilities rather than “raw” probabilities helps us avoid numerical under/overflow.
- E. When doing maximum-likelihood fitting, the parameters with the highest log-likelihood are the same as the parameters with the highest “raw” likelihood.

[5 marks]

**Q14.** REDACTED

**Q15.** For the data in the table, fit a model of the form  $\hat{y} = w_1 + w_2x$

| x | y    |
|---|------|
| 0 | -4.2 |
| 1 | -2.3 |
| 2 | -0.1 |
| 3 | 2.1  |
| 4 | 3.9  |

[5 marks]

- A.  $w_1 = -4.20$ ,  $w_2 = 2.12$ .
- B.  $w_1 = -4.24$ ,  $w_2 = 2.06$ .
- C.  $w_1 = -4.30$ ,  $w_2 = 2.12$ .
- D.  $w_1 = -4.32$ ,  $w_2 = 2.06$ .
- E.  $w_1 = -4.35$ ,  $w_2 = 2.12$ .

**Q16.** For the data in the table, fit a model of the form  $\hat{y} = w_1x + w_2x^2$

| x | y    |
|---|------|
| 0 | -4.2 |
| 1 | -2.3 |
| 2 | -0.1 |
| 3 | 2.1  |
| 4 | 3.9  |

[5 marks]

- A.  $w_1 = -1.42$ ,  $w_2 = 0.603$ .
- B.  $w_1 = -1.55$ ,  $w_2 = 0.654$ .
- C.  $w_1 = -1.60$ ,  $w_2 = 0.674$ .
- D.  $w_1 = -1.78$ ,  $w_2 = 0.687$ .
- E.  $w_1 = -1.82$ ,  $w_2 = 0.742$ .

**Q17.** For the data in the table, fit a model of the form  $\hat{y}_i = w_1X_{i1} + w_2X_{i2}$

| $X_{i1}$ | $X_{i2}$ | $y_i$ |
|----------|----------|-------|
| -1       | -1       | -2.6  |
| -1       | 1        | -0.2  |
| 1        | -1       | 0.5   |
| 1        | 1        | 2.4   |

[5 marks]

(cont.)

- A.  $w_1 = 1.275, w_2 = 0.925$
- B.  $w_1 = 1.305, w_2 = 0.970$
- C.  $w_1 = 1.350, w_2 = 1.025$
- D.  $w_1 = 1.400, w_2 = 1.505$
- E.  $w_1 = 1.425, w_2 = 1.075$

**Q18.** REDACTED

**Q19.** We have  $N$  datapoints,  $x_1, \dots, x_N$ , distributed according to,

$$P(x_i|\mu) \propto \frac{1}{x_i} e^{-(\log x_i - \mu)^2/2}$$

What is the maximum-likelihood solution for  $\mu$ ?

- A.  $\mu = \frac{1}{N} \sum_{i=1}^N x_i$ .
- B.  $\mu = \frac{1}{2N} \sum_{i=1}^N \log x_i$
- C.  $\mu = \frac{1}{N} \sum_{i=1}^N \log x_i$
- D.  $\mu = \frac{1}{2N} \sum_{i=1}^N e^{x_i}$
- E.  $\mu = \frac{1}{N} \sum_{i=1}^N e^{x_i}$

[5 marks]

**Q20.** We have  $N$  datapoints,  $x_1, \dots, x_N$ , distributed according to,

$$P(x_i|\beta) \propto \beta^2 \frac{1}{x_i^3} e^{-\beta/x_i}$$

What is the maximum-likelihood solution for  $\beta$ ?

- A.  $\frac{1}{2} \left( \frac{1}{N} \sum_{i=1}^N x_i \right)$
- B.  $\frac{1/2}{\frac{1}{N} \sum_{i=1}^N x_i}$
- C.  $\frac{1/2}{\frac{1}{N} \sum_{i=1}^N (1/x_i)}$
- D.  $\frac{2}{\frac{1}{N} \sum_{i=1}^N x_i}$
- E.  $\frac{2}{\frac{1}{N} \sum_{i=1}^N (1/x_i)}$

[5 marks]